

TULA, Russian Federation

WMO: 277190

Lat: 54.2335N

Lon: 37.6135E

Elev: 674

StdP: 14.34

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-5.7	-16.6	2.3	-11.3	-11.0	3.1	-4.8	19.4	27.8	16.0	29.3	2.3	10	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	86.3	68.0	82.7	66.5	79.8	65.4	70.1	81.6	68.5	79.5	67.0	77.0	5.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	99.3	75.3	64.7	93.8	73.6	63.1	88.7	72.1	34.4	82.1	33.1	79.5	31.8	77.4	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.2	11.6	DB	-19.4	89.6	6.0	4.3	-23.7	92.8	-27.2	95.3	-30.6	97.7	-34.9	100.9	(4)
(5)			WB	-19.7	72.9	5.8	3.0	-23.9	75.1	-27.3	76.8	-30.5	78.5	-34.7	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	18.6	19.4	28.8	44.1	56.4	62.8	67.5	63.6	53.6	42.6	30.9	22.8
	DBStd	19.54	11.75	12.29	8.61	8.91	8.67	6.56	6.05	6.68	7.20	7.66	9.55	11.29
	HDD50	4456	974	858	656	221	36	1	0	0	43	251	572	844
	HDD65	8393	1439	1278	1121	626	289	119	42	107	346	695	1022	1309
	CDD50	1803	0	0	0	45	236	386	543	420	151	21	1	0
	CDD65	265	0	0	0	0	23	54	121	62	5	0	0	0
	CDH74	2444	0	0	0	7	252	474	1059	610	42	0	0	0
	CDH80	674	0	0	0	1	49	106	315	201	2	0	0	0
Wind	WSAvg	5.2	5.9	6.0	6.0	5.4	4.9	4.5	3.8	4.0	4.1	5.4	6.0	6.1
Precipitation	PrecAvg	24.6	1.7	1.5	1.3	1.7	2.0	2.8	3.0	2.6	2.4	2.2	1.9	1.7
	PrecMax	31.0	3.7	3.1	3.2	3.3	3.4	7.0	6.0	9.4	6.7	4.3	4.4	3.8
	PrecMin	14.7	0.6	0.4	0.5	0.6	0.6	0.9	0.7	0.4	0.3	0.6	0.2	0.7
	PrecStd	3.7	0.7	0.6	0.7	0.7	0.8	1.4	1.3	2.0	1.5	1.1	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	39.2	55.5	74.3	85.1	87.0	92.3	93.4	78.7	67.3	54.1	45.4
		MCWB	37.4	36.0	45.0	55.8	65.9	66.3	68.5	67.7	62.2	55.9	50.5	43.4
	2%	DB	36.0	36.7	47.8	68.3	80.7	83.0	87.0	85.3	74.1	61.1	48.6	40.9
		MCWB	34.5	34.3	39.8	52.7	62.2	66.1	68.2	68.6	61.2	54.4	45.9	39.5
	5%	DB	34.4	35.1	42.9	63.9	76.3	79.7	83.2	80.2	69.7	56.7	45.5	37.2
		MCWB	33.1	33.3	37.4	50.5	60.7	65.1	67.6	65.7	58.8	51.7	43.5	35.9
	10%	DB	32.9	34.0	39.5	59.6	72.0	75.9	80.0	76.4	65.7	53.3	42.5	34.8
		MCWB	31.7	32.6	35.6	48.9	58.4	63.4	66.7	64.0	57.0	49.4	40.9	33.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	36.9	45.6	57.3	67.3	70.4	73.0	71.8	64.6	58.7	50.5	43.8
		MCDB	38.7	38.3	55.8	70.5	82.0	81.8	84.4	83.4	74.6	63.5	53.9	45.2
	2%	WB	34.8	34.9	41.0	54.5	64.1	68.3	70.7	69.3	62.5	55.4	46.5	39.7
		MCDB	35.6	36.0	46.1	64.9	77.2	79.2	81.7	81.4	71.3	59.8	48.0	40.9
	5%	WB	33.2	33.8	38.1	52.3	61.8	66.7	69.2	67.3	60.1	52.3	43.6	36.2
		MCDB	34.1	34.8	41.8	61.8	72.8	76.5	80.1	78.5	67.2	56.1	45.2	37.1
	10%	WB	31.9	32.7	36.0	49.8	59.8	64.7	67.6	65.3	57.9	50.0	41.0	33.8
		MCDB	32.9	33.9	39.0	58.2	69.9	73.3	77.4	74.5	64.5	52.9	42.4	34.6
Mean Daily Temperature Range	5% DB	MDBR	9.5	11.2	12.6	16.9	19.2	17.7	18.1	18.3	16.4	11.7	7.9	8.1
		MCDBR	6.4	7.4	16.9	23.6	24.0	23.2	23.1	24.7	23.2	17.1	11.0	7.4
		MCWBR	6.3	6.4	10.6	11.7	9.9	9.8	8.7	9.9	11.6	11.0	9.3	7.0
	5% WB	MCDBR	6.3	7.1	14.4	20.8	21.7	20.3	20.2	21.6	19.9	14.6	9.7	7.1
		MCWBR	6.3	6.5	9.8	11.6	10.3	9.7	9.3	9.5	10.9	10.6	8.9	6.9
Clear-Sky Solar Irradiance	taub	0.289	0.332	0.350	0.422	0.392	0.383	0.422	0.434	0.387	0.357	0.327	0.293	
	taud	2.118	2.048	2.095	2.105	2.302	2.344	2.259	2.212	2.320	2.338	2.234	2.131	
	Ebn at Noon	201	229	257	253	271	274	260	248	245	225	187	172	
	Edh at Noon	21	32	38	44	38	37	40	39	31	24	19	17	
All-Sky Solar Radiation	RadAvg	235	531	1009	1302	1739	1810	1750	1459	932	451	184	134	
	RadStd	46	62	118	108	145	181	165	154	103	74	35	38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.44	N/A	N/A	N/A	N/A	N/A	N/A	-454	+205	N/A
(47) Regional (11 neighbors)		+1.39	N/A	N/A	N/A	N/A	N/A	-263	-398	+203	+68

Nomenclature: See separate page